



# AN OUTLIER IS **NOT** AUTOMATICALLY AN ERROR.

It could be bad data, a rare event, or your earliest warning signal.

*Investigate. Don't ignore.*

Outliers often tell the most important stories.

## DATA ERROR



e.g., a value entered incorrectly (850 instead of 85).

## RARE BUT REAL EVENT



e.g., a disease outbreak, a successful campaign, or a stock surge.

## EARLY-WARNING SIGNAL



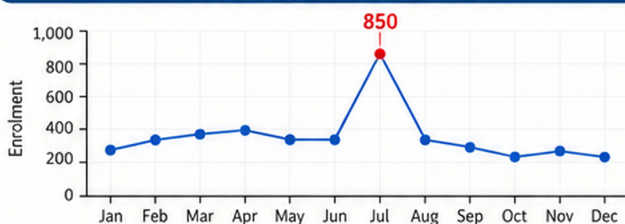
e.g., an unexpected increase in cases, drop in sales, or system failure.

## ACTIONABLE INSIGHT



Verify, understand the cause, and use it to inform decisions and monitoring.

## A REAL-WORLD EXAMPLE Monthly Programme Enrolment

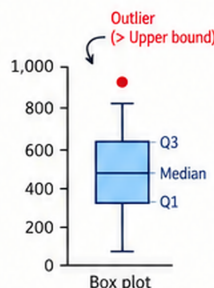


July shows 850 enrolments, much higher than other months.  
**Is this a data error? A successful campaign? Or the start of a new trend?**

Don't remove it blindly. Investigate and understand the context.

## HOW TO DETECT OUTLIERS USING THE IQR METHOD

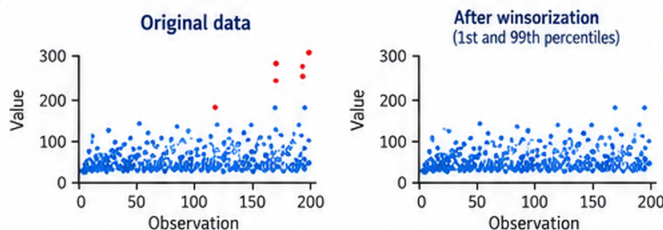
- 1 Calculate Q1 and Q3**  
(Q1 = 25th percentile, Q3 = 75th percentile)
- 2 Compute the interquartile range (IQR)**  
 $IQR = Q3 - Q1$
- 3 Define the outlier thresholds**  
Lower bound =  $Q1 - 1.5 \times IQR$   
Upper bound =  $Q3 + 1.5 \times IQR$
- 4 Flag values outside these bounds**  
as potential outliers (not automatically errors).



- ✓ The IQR method is simple, robust, and widely used. It works well for many real-world datasets.

## OPTIONAL TREATMENT: WINSORIZATION

Winsorization limits extreme values by capping them at a chosen percentile (e.g., 1st and 99th). It reduces the influence of outliers without removing data.



Winsorization is a tool, not a rule. Always understand the cause of outliers and consider whether to investigate, model separately, transform, or winsorize.

## USE CASES BEYOND HEALTH



### Public Health

Sudden increase in disease cases, unusual dropout rates, or unexpected vaccine demand.



### Business & Sales

Unusual spike or drop in monthly sales, website traffic, or customer sign-ups.



### Operations

Anomalies in system logs, equipment failure, or supply chain delays.



### Finance

Fraud detection, unusual transactions, or market shocks.



### Environment & Climate

Extreme weather events, unusual temperature or rainfall patterns.

## THE EVIDENCELAB WORKFLOW FOR OUTLIERS



## KEY TAKEAWAYS

- ✓ Outliers are not always errors — they can be valuable information.
- ✓ Use the IQR method to detect potential outliers.
- ✓ Winsorization can reduce the influence of extreme values.
- ✓ Always investigate and understand the context.
- ✓ Outliers and anomalies can provide early-warning signals.
- ✓ The same methods apply in health, business, finance and beyond.



## COMMON MISTAKES

- Removing outliers without understanding why.
- Assuming all outliers are data errors.
- Using arbitrary cut-offs.
- Ignoring outliers in time-series monitoring.
- Failing to document your decisions.



## TRY IT YOURSELF!

- ✓ Open the Google Colab notebook.
- ✓ Run all cells (no setup required).
- ✓ Explore real-world examples.
- ✓ Try IQR detection and winsorization.
- ✓ Apply anomaly detection to time-series data.



Open in Colab

Scan the QR code or click the link.

